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Concordance Correlation Coefficient (CCC) — 一致性相关系数

1) **What it is (定义)**

The **Concordance Correlation Coefficient (CCC)**, proposed by **Lin (1989)**, measures how well two measurements **agree** with each other—not just whether they're correlated, but whether they fall close to the **45° line of perfect agreement** (i.e., $y = x$).

Formula:

$$\mathrm{CCC} = \frac{2\rho\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2+(\mu_x-\mu_y)^2}$$

where:

- ρ = Pearson correlation between x and y
- σ_x, σ_y = standard deviations
- μ_x, μ_y = means

Range: CCC ∈ [−1, 1], where **1 = perfect concordance**, 0 = no agreement, −1 = perfect disagreement.

2) **CCC vs Pearson correlation (关键区别)**

Metric	What it measures	Sensitive to…
Pearson r	Linear **association** (strength & direction)	Only the **shape** of the relationship (not location/scale)

Metric	What it measures	Sensitive to…
CCC	**Agreement** (how close points are to $y = x$)	Both correlation **and** systematic bias (mean/variance differences)

Example:

- If method A always gives values 10 units higher than method B ($y = x + 10$), Pearson $r = 1.0$ (perfect correlation), but **CCC < 1** (poor agreement because of the bias).

3) **When to use CCC (应用场景)**

CCC is the standard metric for **method comparison / inter-rater reliability / test-retest studies** where you care about **absolute agreement**, such as:

- **Comparing two measurement devices** (e.g., manual vs automated segmentation volumes)
- **Inter-rater reliability** (two raters measuring the same thing)
- **Test-retest reliability** (same method on the same subjects at two time points)
- **Algorithm validation** (e.g., does a new MRI pipeline give the *same numbers* as FreeSurfer?)

Why not just use Pearson r? Because high correlation doesn't guarantee the methods give similar *absolute* values—CCC penalizes both **bias** (mean shift) and **scale differences** (variance mismatch).

4) **Interpretation guidelines (经验阈值)**

There's no universal cutoff, but common conventions (adapted from various fields):

CCC	Interpretation
> 0.95	Excellent agreement
0.90–0.95	Substantial agreement
0.75–0.90	Moderate agreement
< 0.75	Poor agreement

(These are rough; context matters—e.g., clinical vs research use.)

5) **Relation to other agreement metrics**

- **Intraclass Correlation Coefficient (ICC)**: closely related; ICC(2,1) or ICC(3,1) often numerically very similar to CCC in balanced designs.
- **Bland-Altman plot**: visualizes agreement (bias & limits of agreement); CCC is a single summary number, Bland-Altman shows the pattern.
- **Mean Absolute Error (MAE) / RMSE**: measure magnitude of disagreement but don't account for correlation structure.

Best practice: report CCC **and** show a Bland-Altman plot (or scatter plot with $y = x$ line) to give both a summary metric and visual context.

6) **Example in neuroimaging context**

Suppose you compare hippocampal volumes from **FreeSurfer** vs **FastSurfer** on 100 subjects:

- **Pearson** $r = 0.98$: strong linear relationship ✓
- **CCC = 0.85$$**: moderate agreement—suggests there's some **systematic bias or scale difference** even though the correlation is high.

→ You'd then check: does FastSurfer systematically over/underestimate? (mean difference) and is the variance similar? (Bland-Altman would show

this visually.)

Quick summary (TL;DR)

- **CCC** = measures **agreement** (not just correlation); penalizes bias and scale differences.
- **Use it** when you care whether two methods give the **same absolute numbers** (method comparison, reliability).